AUTOMATIC HTML CODE GENERATOR FROM MOCKUP IMAGES USING MACHINE LEARNING TECHNIQUES

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Abstract: Every website begins with the designing phase. Firstly, the design of the website should be drafted either with the help of designing tools or hand drawn. Once the design phase is completed code writing process begins which involves several breakdown steps of instructions. In our paper, code writing process is eliminated and only hand drawn image of the website prototype is uploaded as an input to the algorithm. This hand drawn image is processed through a series of repetitive steps which works on the principle of neural networks and identifies the components in the image. The components and text are detected using OCR technique. Once components are detected they are cropped and differentiated and find. the suitable match from the training data set using deep neural networks CNN technique. Once the components are detected and analyzed it will generate HTML code for the mockup image. This process is repeated several times until the desired result is obtained. During this process several images are produced as output during preprocessing stage. These images are obtained by processing single image through multiple phases using neural networks. Output of one phase is given as an input to the next phase during processing of an image. This multiple phase processing will produce accurate result.

Index terms - OCR, Deep Neural Networks, CNN, Mockup Image, Components.

INTRODUCTION

The importance of the websites has significantly increased due to the progress in present technology. Nowadays all the institutions, communities, people etc. are using websites to reflect their identity. Companies utilize these websites for marketing and advertising purposes. On the other hand, educational institutions, communities use websites to represent their services. At the front end of the web site web page is the main component which provides an interaction to the users. So, the design of the web page must be simple, interactive and efficient with wide range of flexible options. It is crucial to serve a page that is attractive to the end user and easy to use. However, developing webpage that responds efficiently to the users requires critical task in coding. In the process of developing code many different teams like specialists, software designers, graphic designers need to coordinate which is time consuming process. So, to avoid this complete process we proposed an idea in which user will make use of mockup images to generate HTML code. This way users can get the webpage instantly with minimal errors and desired design. By this process repetitive tasks by different teams and spending more time in debugging is reduced.

RELATED WORK

In the current advancing technological world, web pages role has become prominent. Many researchers provided different approaches to generate HTML code easily using different algorithms and principles. [1] In Factors Involved in Artificial Intelligence-based Automated HTML Code Generation Tool with the help of advanced technologies like artificial intelligence and machine learning developed continuous upgrade in the overall performance of converting mockup image to code by following 13 steps which includes from uploading mockup image to conversion of that to image format followed by OCR, object captioning, image processing, payloads etc. [2] In HTML code generation using CNN algorithm code is generated by a process which involves detection, dilation, erosion, classification and assembling of image. [3] In Automatic HTML Code Generation from Mock-up Images Using Machine Learning Techniques it involves object detection, cropping and object recognition to generate code from the mockup image. HTML builder in this model will develop html code as an output file. HTML builder will work based on the contour detection of the components. The proposed algorithm is shown in the below figure 2.1.

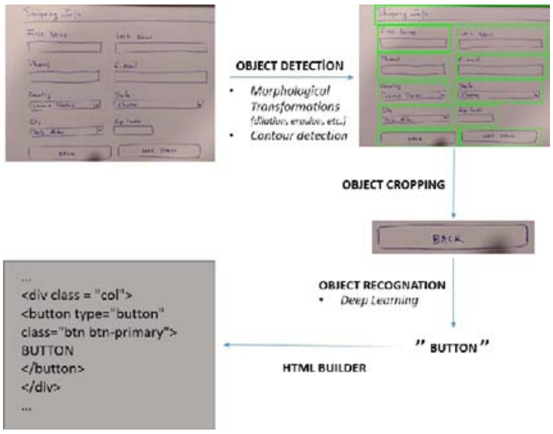

Figure 2.1 Proposed algorithm of one of the existing prototype

PROPOSED SYSTEM

In our proposed system, to provide better interactive interface user should create an account and login to the application with respective details. On successful login, user needs to go through series of steps to get html code for the mockup image. Firstly, user needs to ensure the Wamp server is online or not because it provides localhost user to host the application. Then user should upload mockup image and submit it. As a next step, which is image preprocessing user must upload the same mockup as in the previous step here. After submitting the image in second step it results in showing multiple images like gray scale, erosion, threshold image which are obtained by passing the mockup image to deep learning neural networks. After this phase user should upload the same image to again to obtain the html file. So, this complete process includes detection of components, cropping of components and recognition components. The model is trained by predefined dataset.

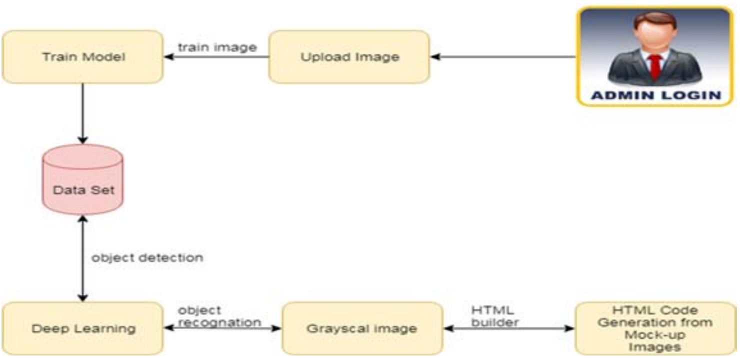

Figure 3.1: Architecture diagram of proposed system

ADVANTAGES OF PROPOSED SYSTEM

i. It requires less human support.
ii. Minimal coding.
iii. It produces efficient outcomes.
iv. Code for static webpages is obtained quickly.
v. Less time consuming.
vi. Easy to use.

WORKING OF PROPOSED SYSTEM

Algorithm

For the Algorithmic approach towards the proposed system, multiple principles and techniques are followed.

1. OCR (Optical Character Recognition) - It is used for recognizing the characters in the mockup image.
2. REMAUI (Reverse Engineering Mobile Application User Interface) - This process is considered as REMAUI because it produces code from image which is vice versa to the actual process.
3. Deep Neural Networks - Neural networks are used to process image through different layers so as to identify the components.

Steps to obtain html code from mockup image:

1. Login via new user registration
2. Upload Mockup image
3. Upload image for Mockup preprocessing phase
4. Upload image for processing phase
5. HTML file as an output

Login via new user registration

In this step, new user needs to register into the database to login to the application. Upon creation of new account user will be able to login. This includes details like username, emailed, password. Below shown figures are details of registration page and login page respectively.

Figure 5.1: Registration Page

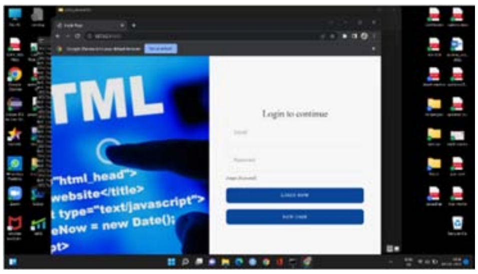

Figure 5.2: Login Page

Upload Mockup Image

In this step user should upload the mockup image which is of type .PNG. Once the image is uploaded submit it to the application. After submission it user OCR technique to recognize the text components and characters.

Figure 5.3: Upload Mockup Image

Upload image for Mockup Preprocessing Phase

In this phase user should upload the same mockup image as previous. After uploading the image, it produces multiple images like gray scale image, threshold image, erosion image etc. All these images are produced through several layers of processing. Each layer will produce one image as output. The output of first layer is sent as an input to the next phase until the last layer. Use of this phase it recognizes all the components and crop the patterns for each component.

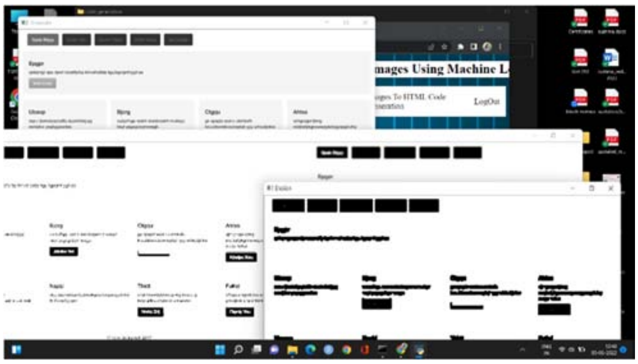

Figure 5.4 Output images of neural network layers

Upload Image of processing phase

In this step, user should upload the same mockup image again. This phase output will be the html file URL which need to be executed in any editor to crosscheck the code. We can use any of the web browser to run the file.

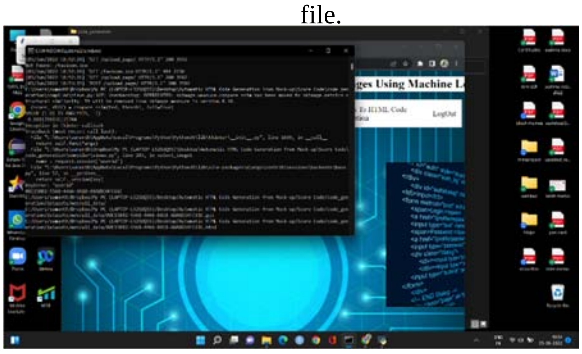

Figure 5.5 HTML file URL HTML file as an output

The code URL which is generated in the previous image is made to run in any of the web browser. It produces code for the uploaded mockup image. It can be verified by running the script file in any of the editor.

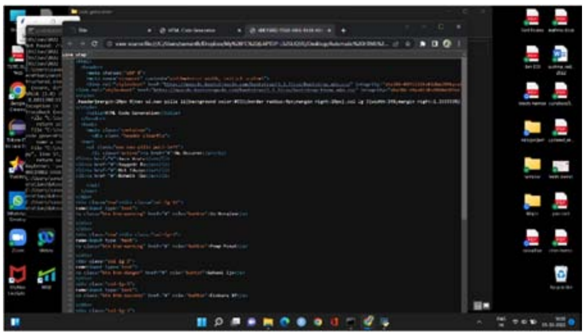

Figure 5.6 HTML Script file

RESULT

Below are few captured images of the model.

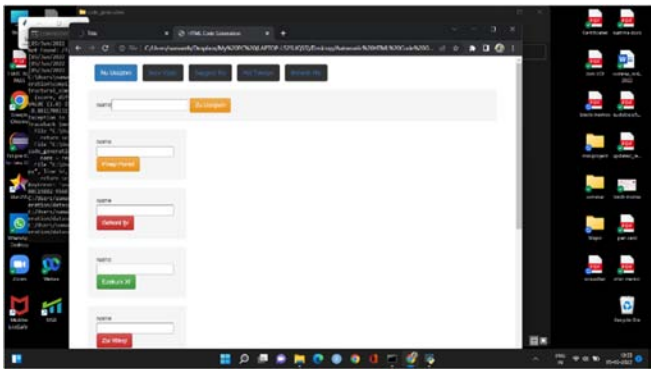

Figure 5.7: Output of the HTML file verified in a web browser

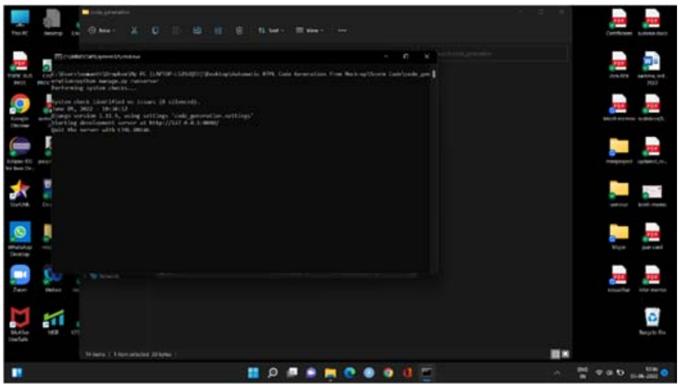

Figure 5.8 Virtual Environment

Performance evaluation

Table 6.1.1: Performance Criteria of various algorithms.

Performance Criteria	OCR	Neural Networks	OCR + Neural Networks
Detection	**90%**	**94%**	**96%**
Cropping	**96%**	**92%**	**96%**
Dilation	**99%**	**97.4%**	**98%**

CONCLUSION

This model will generate HTML code using different techniques like OCR, Neural Networks, Image processing which assists users to generate HTML code instantly without much effort. It produces less coding errors which ensures developers to concentrate on logical process of the website rather than templates. A basic static design can be obtained from which further modifications can be made to obtain user specified web page.

FUTURE SCOPE

This model does not support for animations, videos and series of images. Moreover, to obtain a HTML code for the mockup image there should be an associated GUI file for every image. So, system can be enhanced in such a way that it supports animations to produce dynamic web pages without GUI file in the system. The approach of enhancing system requires is more dataset with different models and examples. By providing huge amount of dataset system can be improved to make dynamic pages with different hover components, media queries and alternatives.

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